

# 3.3.7 Lesson Review

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Passing Score: 80%



### Question 1

✓ Correct

During prompt and log monitoring for an AI system, which of the following are typically logged to support operational monitoring and regulatory compliance? (Select three.)

- ☐ Operating system version on user devices
- ☒ System latency or response time measurements ✓ Correct
- ☐ Detailed user mouse movements within the application
- ☐ Network packet loss rates for all connected devices
- ☐ CPU temperature statistics for the server hardware
- ☒ Model responses generated for users ✓ Correct
- ☒ Input prompts submitted to the AI system ✓ Correct

### Explanation

Input prompts submitted to the AI system is a correct answer. Input prompts are logged to monitor what users are asking the AI system. This helps in tracking for potentially restricted inputs, injection attempts, and monitoring for compliance with organizational policies.

Model responses generated for users is a correct answer. Logging model responses is essential for auditing what information the AI system provides to users. This supports data loss prevention efforts and ensures that the system is not releasing sensitive or inappropriate content.

System latency or response time measurements is a correct answer. Tracking system latency (how long the system takes to respond) helps identify performance bottlenecks. It ensures the system can handle workload demands efficiently and alerts administrators to potential operational issues.

CPU temperature statistics for the server hardware is incorrect. CPU temperature is a technical hardware metric not directly related to prompt or log monitoring for AI system inputs, outputs, or audit purposes. While useful elsewhere, it is outside the scope of AI-specific monitoring.

Operating system version on user devices is incorrect. While OS version might be important in endpoint management or compatibility audits, it is not a standard item logged by AI prompt or log monitoring processes.

Detailed user mouse movements within the application is incorrect. Tracking user mouse movements is more relevant to UI/UX analytics, not to AI system prompt and log monitoring—where the focus is on transactional and compliance-related data.

Network packet loss rates for all connected devices is incorrect. Packet loss monitoring is related to general network health, not specific to the prompts, responses, or compliance logs managed by an AI system.

### **Related Content**



#### **3.3.1 Prompt and Log Monitoring**

resources\questions\q\_prompt\_and\_log\_monitoring\_03.question.xml

An AI-powered knowledge base system provides technical answers to users based on a mix of internal company documents and internet-sourced data. Recently, some users complained that responses, while confidently delivered, were sometimes incorrect or out-of-date.

As an auditor focused on quality and compliance, you must apply appropriate techniques to ensure that the system's answers are factually correct.

What is the FIRST step you should take to address this issue in line with quality and compliance auditing practices?

- ☐ Increase the frequency of system restarts and purge the cache regularly to eliminate outdated information that may be influencing responses, then notify IT staff to monitor changes.
- ☐ Optimize log retention settings to maximize available storage for prompts and responses, and set up a deduplication process to keep only unique knowledge base entries for audit efficiency.
- ☐ Request that users escalate incorrect responses to a manual review team, create a policy for retroactive correction of all previously delivered answers, and require weekly retraining.
- ☒ Implement grounding checks to verify that AI-generated answers are tied back to trusted, up-to-date data sources, ensuring responses are factually accurate. ✓ Correct

### Explanation

Implementing grounding checks to verify that AI-generated answers are tied back to trusted, up-to-date data sources, ensuring responses are factually accurate, is the correct answer. Implementing grounding checks ensures that each response can be traced back to an accurate, trusted source, which directly addresses the problem of factual inaccuracies in AI-generated content. This aligns closely with best practices.

Increasing the frequency of system restarts and purging the cache regularly to eliminate outdated information that may be influencing responses, then notifying IT staff to monitor changes, is incorrect. Regular system restarts and purging the cache may help clear some outdated information from memory but do not directly ensure responses are fact-checked or corrected. These technical actions lack a targeted audit approach for validating answers.

Requesting that users escalate incorrect responses to a manual review team, creating a policy for retroactive correction of all previously delivered answers, and requiring weekly retraining is incorrect. This response is reactive and resource-intensive. It doesn't provide a systematic, immediate method to ensure that all future AI responses are grounded in fact, nor does it use automated techniques for ongoing quality assurance.

Optimizing log retention settings to maximize available storage for prompts and responses, and setting up a deduplication process to keep only unique knowledge base entries for audit efficiency, is incorrect. Maximizing log storage and deduplicating entries do not help identify or correct factually inaccurate information provided to users. These steps focus on data management, not the content's truthfulness or its verification against trusted sources.

#### **Related Content**



#### **3.3.4 Quality and Compliance Auditing**

resources\questions\q\_quality\_and\_compliance\_auditing\_05.question.xml

### Question 3

✓ Correct

Which of the following is a recommended action if prompt and log monitoring detect model drift in an AI system?

- ☐ Temporarily narrow the model's scope of use without adjusting the training data
- ☒ Retrain the model to re-baseline its performance ✓ Correct
- ☐ Apply manual rule-based overrides to compensate for the drift
- ☐ Increase the model's confidence threshold so fewer predictions are returned

### Explanation

Retraining the model to re-baseline its performance is the correct answer. When model drift is detected, the recommended action is to retrain the model using updated, higher-quality, or more representative datasets. This process recalibrates the model so that its predictions align with current data patterns and business requirements, effectively re-baselining performance. Retraining helps restore accuracy, relevance, and reliability, and it can be tracked and validated as part of a formal MLOps or AI governance process.

Applying manual rule-based overrides to compensate for the drift is incorrect. Introducing manual rules or "patches" on top of a drifting model may appear to fix issues quickly, but it does not address the underlying problem: the model's learned patterns no longer match real-world data. Over time, rule-based overrides can become complex, brittle, and hard to maintain, masking deeper accuracy and bias issues. Best practice is to retrain or update the model so its statistical behavior itself is corrected.

Temporarily narrowing the model's scope of use without adjusting the training data is incorrect. Limiting where or how a model is used can reduce immediate risk, but it does not resolve model drift. The model will continue to be misaligned with the evolving data environment, and as usage expands again, the same or worse performance problems will reappear. Scope restriction can be a short-term risk control, but recommended practice is to retrain or recalibrate the model based on current data, not simply constrain its deployment.

Increasing the model's confidence threshold so fewer predictions are returned is incorrect. Raising the confidence threshold can make the system appear safer but a drifting model can still be confidently wrong if its internal representation is outdated or misaligned. Adjusting thresholds is a tuning step, not a remediation for drift. When monitoring shows true model drift, the underlying model should be retrained or updated to restore trustworthy performance rather than merely filtering its outputs more aggressively.

#### **Related Content**



##### **3.3.1 Prompt and Log Monitoring**

resources\questions\q\_prompt\_and\_log\_monitoring\_04.question.xml

A healthcare organization uses an AI system to assist clinicians by providing medication recommendations. As part of your role in quality and compliance auditing, you notice some AI-generated responses sound uncertain or vague, while others are precise and confident.

To improve reliability and safety, you decide to apply response confidence level monitoring as a recommended best practice.

Which action should you take FIRST to apply response confidence level monitoring in line with quality and compliance standards?

- ☐ Audit access logs to ensure only authorized personnel are using the AI system, and set up automated alerts for unusual usage patterns or changes in system configuration that could impact decision accuracy.
- ☐ Increase the frequency of retraining for the AI system and require a periodic manual audit of all medication recommendations made over the last quarter to spot inconsistencies or errors.
- ☐ Archive all AI-generated recommendations and responses for five years, implementing strict role-based access control to ensure data integrity and compliance with healthcare regulations.
- ☒ Configure the AI system to tag each recommendation with a confidence score, allowing clinicians and auditors to review and act on recommendations with low confidence levels. ✓ Correct

### Explanation



Configuring the AI system to tag each recommendation with a confidence score, allowing clinicians and auditors to review and act on recommendations with low confidence levels, is the correct answer. Tagging each response with a confidence score aligns directly with response confidence level monitoring. This approach enables clinicians and auditors to identify and appropriately handle recommendations with low confidence, improving decision-making safety and facilitating targeted auditing.

Increasing the frequency of retraining for the AI system and requiring a periodic manual audit of all medication recommendations made over the last quarter to spot inconsistencies or errors is incorrect. These actions do not apply confidence monitoring to individual AI responses. This is resource-intensive and retrospective, not a proactive solution for real-time confidence assessment.

Auditing access logs to ensure only authorized personnel are using the AI system, and setting up automated alerts for unusual usage patterns or changes in system configuration that could impact decision accuracy, is incorrect. This does not relate to tracking or acting on the confidence level of individual AI recommendations. These actions help safeguard operations but do not provide insights into the quality or certainty of AI-generated responses.

Archiving all AI-generated recommendations and responses for five years, implementing strict role-based access control to ensure data integrity and compliance with healthcare regulations, is incorrect. These are important for compliance, especially in healthcare, but do not involve monitoring or using response confidence levels to enhance the reliability of current recommendations provided to clinicians.

#### **Related Content**



#### **3.3.4 Quality and Compliance Auditing**

resources\questions\q\_quality\_and\_compliance\_auditing\_06.question.xml

### Question 5

✓ Correct

Which of the following best explains how adjusting the rebuild frequency of an AI system can impact its operation?

- ☐ Decreasing rebuild frequency guarantees faster response times for the AI system
- ☐ Decreasing rebuild frequency eliminates the need for compliance audits
- ☒ Increasing rebuild frequency can help reduce hallucinations and bias by incorporating new, accurate data ✓ Correct data
- ☐ Increasing rebuild frequency allows unlimited storage of historical data

### Explanation

Increasing rebuild frequency can help reduce hallucinations and bias by incorporating new, accurate data is the correct answer. More frequent rebuilds (retraining or updating) help address issues like hallucinations and bias by integrating current, higher-quality data.

Decreasing rebuild frequency guarantees faster response times for the AI system is incorrect. Faster response times are not a guaranteed result of rebuild frequency; this depends more on hardware and configurations.

Increasing rebuild frequency allows unlimited storage of historical data is incorrect. Rebuild frequency does not affect the system's historical data storage limits.

Decreasing rebuild frequency eliminates the need for compliance audits is incorrect. Compliance audits remain necessary regardless of how often the system is rebuilt.

### Related Content



3.3.2 Performance and Cost Monitoring

resources\questions\q\_performance\_and\_cost\_monitoring\_04.question.xml

Your company's AI-powered loan approval system has shown a pattern where applicants from certain demographic groups receive disproportionately fewer approvals, even when their financial profiles are similar to those of other applicants.

As a compliance auditor, you are tasked with applying quality and compliance auditing practices to address concerns regarding fairness in the AI system.

Which of the following actions should you take FIRST to evaluate and address fairness issues in the AI system?

- ☐ Retrain the AI model using a golden dataset and immediately implement new policies that require manual review of all loan rejections while extending system access logs to cover all model predictions.
- ☐ Focus on auditing all activity logs for suspicious administrative access and log manipulation to ensure that elevated permissions have not led to system bias, followed by encryption of logs.
- ☒ Analyze the AI system's outputs using counterfactual testing, altering applicant demographic variables to check if the output changes and thereby identify instances of bias or unfairness in decision-making. ✔ Correct
- ☐ Increase storage capacity for the AI system so that a greater volume of applicant data can be retained, enabling future audits, and regularly prune logs using an LRU eviction system to optimize performance.

### Explanation

Analyzing the AI system's outputs using counterfactual testing, altering applicant demographic variables to check if the output changes, and thereby identifying instances of bias or unfairness in decision-making is the correct answer. Counterfactual testing directly addresses fairness by systematically altering individual demographic variables in the data to observe if the AI decision changes. This can pinpoint bias in the model's predictions and is an efficient, targeted way to audit for fairness issues.

Retraining the AI model using a golden dataset and immediately implementing new policies that require manual review of all loan rejections while extending system access logs to cover all model predictions is incorrect. This may help in the long term, but these actions are not the initial or most focused response for evaluating present fairness issues. Extending system access logs is more about tracking user behavior and security, not directly diagnosing fairness in automated decisions. This mixes procedural changes with AI modification without first isolating the problem.

Focusing on auditing all activity logs for suspicious administrative access and logging manipulation to ensure that elevated permissions have not led to system bias, followed by encryption of logs, is incorrect. Encrypting logs is also a security-focused action. These steps do not apply AI fairness evaluation techniques covered in the uploaded material.

Increasing storage capacity for the AI system so that a greater volume of applicant data can be retained, enabling future audits, and regularly pruning logs is incorrect. Log pruning and storage optimization are unrelated to the identification or remediation of bias or fairness issues in the AI model's outputs.

#### **Related Content**



3.3.4 Quality and Compliance Auditing

resources\questions\q\_quality\_and\_compliance\_auditing\_04.question.xml

When monitoring AI system responses, what is a primary reason for evaluating the accuracy of those responses?

- ☐ To allow users to bypass security restrictions
- ☒ To identify when the system may require retraining for improved performance ✓ Correct
- ☐ To increase the number of response tokens used by the AI system
- ☐ To ensure users are submitting prompts in the correct language

### Explanation

To identify when the system may require retraining for improved performance is the correct answer. Monitoring response accuracy helps organizations detect if the AI model is producing inaccurate or low-quality outputs, which may signal issues like model drift and the need for retraining to restore or improve performance.

To ensure users are submitting prompts in the correct language is incorrect. While prompt language is important, response monitoring is not primarily focused on verifying prompt language but rather on the quality and accuracy of the AI's outputs.

To increase the number of response tokens used by the AI system is incorrect. The goal of monitoring responses is to ensure quality and accuracy, not to increase token use, which could actually increase costs unnecessarily.

To allow users to bypass security restrictions is incorrect. Security should not be compromised; monitoring aims to enforce, not bypass, policies and restrictions.

### Related Content



3.3.1 Prompt and Log Monitoring

resources\questions\q\_prompt\_and\_log\_monitoring\_02.question.xml

A company has deployed an AI system that processes thousands of customer queries daily. Recently, users have reported increased response times and inconsistent output quality, especially during peak hours.

As the system administrator, you are tasked with improving the system's performance and ensuring cost-effectiveness.

Which of the following adjustments should you prioritize based on best performance monitoring practices described in the reading?

- ☐ Limit access to system logs by setting them to read-only
- ☐ Only audit for hallucinations in the AI system
- ☐ Mask personally identifiable information (PII) in stored prompts
- ☒ Increase hardware resources during peak usage times ✓ Correct

### Explanation

Increase hardware resources during peak usage times is the correct answer. Performance monitoring includes tracking query and response rates to identify peak usage times. Increasing hardware resources during those times directly addresses slow response and degraded output quality, aligning with best practices for optimized operation and ensuring the system can handle peak workloads efficiently.

Limit access to system logs by setting them to read-only is incorrect. Limiting access to logs by setting them to read-only is a measure designed for protecting log integrity, not for improving live system performance or responsiveness during peaks.

Only audit for hallucinations in the AI system is incorrect. Auditing for hallucinations is a quality and compliance activity. While important, it doesn't address the immediate performance issues of slow response time and workload management.

Mask personally identifiable information (PII) in stored prompts is incorrect. Masking PII in stored prompts is a crucial log sanitization process for data privacy, but it does not impact the AI system's capacity to process queries or improve performance during peak hours.

### Related Content

 3.3.2 Performance and Cost Monitoring

Question 9

✓ Correct

Why is prompt compression used as part of performance monitoring in AI systems?

- ☐ To make logs more readable for administrators
- ☐ To increase the frequency of error messages generated
- ☐ To automatically encrypt log files for protection

☒ To reduce the workload and cost in token-limited environments

✓ Correct

**Explanation**

To reduce the workload and cost in token-limited environments is the correct answer. Compression minimizes the computational effort required, which lowers both latency and cost, making the system more efficient.

To make logs more readable for administrators is incorrect. Prompt compression is unrelated to log readability.

To increase the frequency of error messages generated is incorrect. The goal is cost reduction and performance improvement, not generating error messages.

To automatically encrypt log files for protection is incorrect. Encrypting logs is a security practice, not related to prompt compression.

**Related Content**



3.3.2 Performance and Cost Monitoring

resources\questions\q\_performance\_and\_cost\_monitoring\_01.question.xml

What is one potential benefit of using prompt compression in environments where AI system usage is charged by token usage?

- ☐ It guarantees all system responses are more accurate
- ☒ It reduces the number of tokens required, helping to control operational costs ✓ Correct
- ☐ It ensures all user prompts are fully encrypted for privacy
- ☐ It eliminates the need for logging or monitoring AI inputs

### Explanation

It reduces the number of tokens required, helping to control operational costs is the correct answer. Since many AI platforms charge based on tokens, prompt compression directly helps to lower the overall cost by using fewer tokens.

It guarantees all system responses are more accurate is incorrect. Prompt compression does not guarantee accuracy, although it can help responses be more direct.

It ensures all user prompts are fully encrypted for privacy is incorrect. Compression is not a form of encryption for privacy; these are separate processes.

It eliminates the need for logging or monitoring AI inputs is incorrect. Logging and monitoring remain necessary regardless of prompt compression strategies.

### Related Content



3.3.2 Performance and Cost Monitoring

resources\questions\q\_performance\_and\_cost\_monitoring\_03.question.xml